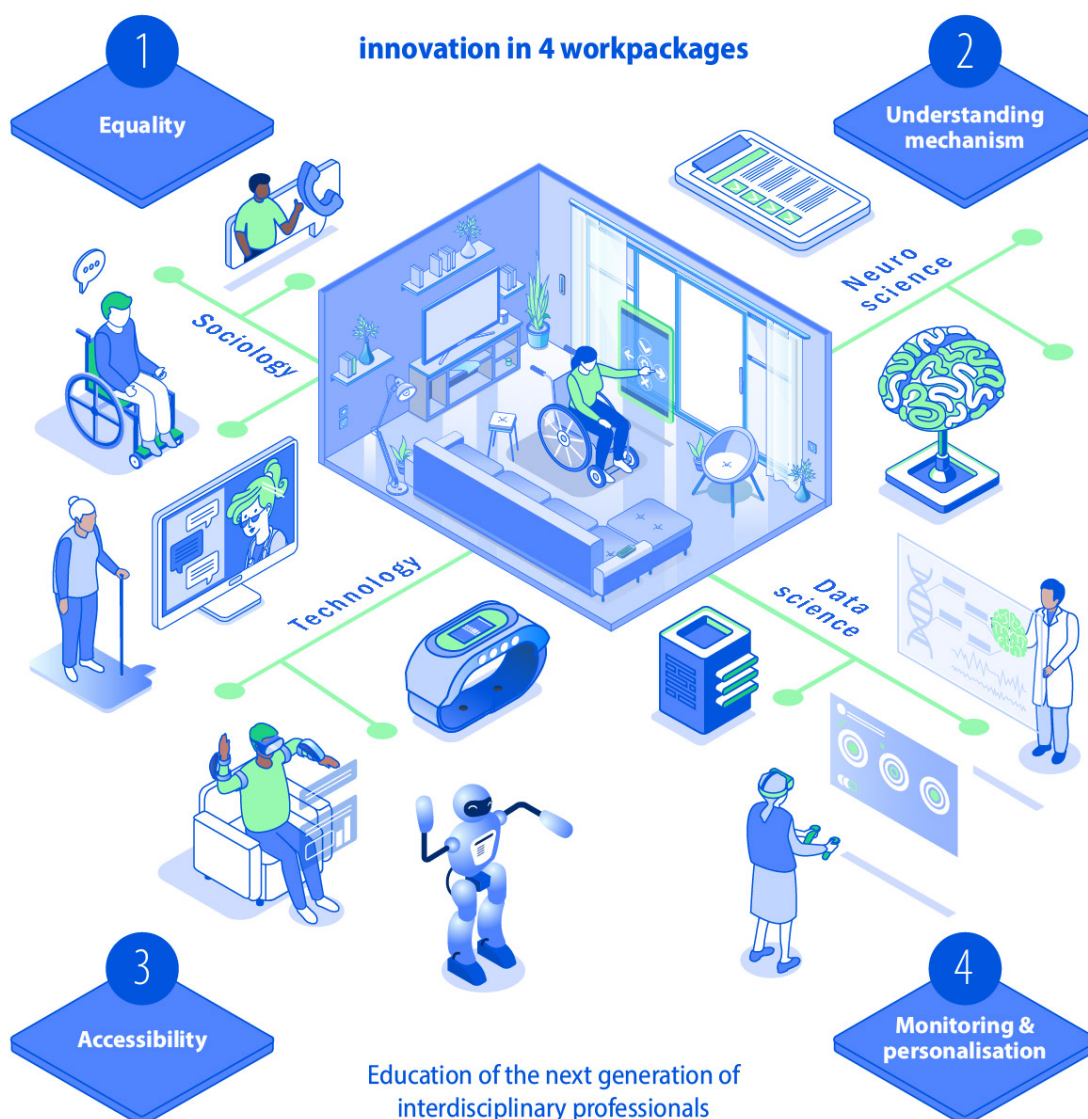


Convergence Human Mobility Center

Tailored technology supports at home treatment

The mission of the Convergence Human Mobility Centre is to enhance the effectiveness and acceptability of minimally supervised rehabilitation with equal access for all



Infographic©Miranda Nell

1. General information

Title:

User-centered technology-based interventions for minimally supervised motor rehabilitation; a **Convergence Human Mobility Center (CHMC) flagship**

1.a Applicants

Name*	Institution	Department	Email	Tenured/Tenure-track/Other	Erasmus MC applicants: mix/clinician/researcher	Role in Flagship
Prof. Gerard Ribbers	Erasmus MC	Rehabilitation Medicine	g.ribbers@erasmusmc.nl	Tenured	mix	Main Lead
Prof. Jane Cramm	EUR	Erasmus School of Health Policy & Management	Cramm@eshpm.eur.nl	Tenured	researcher	Lead applicant WP Leader
Dr. Laura Marchal-Crespo	TU Delft	Cognitive Robotics	L.MarhalCrespo@tudelft.nl	Tenured	researcher	Lead applicant WP Leader
Dr. Agaath M.C. Sluiter	Erasmus MC / Rijndam	Rehabilitation Medicine	a.sluiter@erasmusmc.nl	Tenured	Strategic advisor / Manager innovation Rijndam a.i.	Co-applicant / Program coordinator
Dr. Ajay Seth	TU Delft	Biomechanical engineering	a.seth@tudelft.nl	Tenure track	researcher	Co-applicant WP Leader
Prof. Maarten Frens	Erasmus MC	Neuroscience	m.frens@erasmusmc.nl	Tenured	researcher	Co-applicant WP leader
Prof. Dr. Ing Heike Vallery	TU Delft	Biomechanical engineering	h.vallery@tudelft.nl	Tenured	researcher	Co-applicant
Dr. Hans Bussmann	Erasmus MC	Rehabilitation Medicine	j.b.j.bussmann@erasmusmc.nl	Tenured	researcher	Co-applicant
Prof. Ruud Selles	Erasmus MC	Rehabilitation Medicine	R.selles@erasmusmc.nl	Tenure	researcher	Co-applicant
Dr. Arkady Zgonnikov	TU Delft	Cognitive Robotics	A.Zgonnikov@tudelft.nl	Tenure track	researcher	Co-applicant
Dr. Manon Kok	TU Delft	Center for Systems and Control	m.kok-1@tudelft.nl	Tenure track	researcher	Co-applicant
Dr. Ruben Regterschot	Erasmus MC	Rehabilitation Medicine	g.r.h.regterschot@erasmusmc.nl	Tenure track	researcher	Co-applicant
Dr. Gerwin Smit	TU Delft	Biomechanical engineering	g.smit@tudelft.nl	Tenure track	researcher	Co-applicant
Dr. Erik Grauwmeijer	Erasmus MC / Rijndam	Rehabilitation Medicine	egrauwmeijer@rijndam.nl	Tenured	mix	Co-applicant

Dr. Rutger Osterthun	Erasmus MC / Rijndam	Rehabilitation Medicine	rosterthun@rijndam.nl	Tenured	mix	Co-applicant
Dr. Bram Onneweer	Erasmus MC / Rijndam	Living labs Rijndam	bonneweer@rijndam.nl	Tenured	researcher Living Lab coordinator	Co-applicant
Hans Stravers	Laurens	Geriatric Rehabilitation	h.stravers@laurens.nl	Tenured	na	Co-applicant

Corresponding Lead: Dr. Agaath M.C. Sluijter MBA (organizational program leader)

Planned start date of seed funding: 1-9-2022

Primary Convergence theme: *Human Centered Technology and AI for Health*

Secondary Convergence theme (if applicable): *Technology Supported Transitions in Health(care)*

Tertiary Convergence theme (if applicable): *Improving Health Journeys*

Quaternary Convergence theme (if applicable): *Fundamentals of Health & Disease*

Keywords: *Rehabilitation, motor Impairments, therapy, socio economic status, equality, efficacy, sustainability*

1.b Summary

Demand for healthcare is growing and becoming increasingly complex. Stroke is in the top five of diseases with the highest annual care costs, where patients face numerous obstacles in accessing the necessary care for rehabilitation. Moreover, as society ages, the number of stroke patients is expected to increase. A paradigm shift is needed from conventional on-site labour-intensive motor rehabilitation to minimally supervised motor rehabilitation at the patient's home.

Initiated by partners from TU Delft, Erasmus MC, EUR, Rijndam Rehabilitation and Laurens geriatric rehabilitation, the **Convergence Human Mobility Center (CHMC)** will enhance the effectiveness and acceptability of user-centred technology-based minimally supervised motor rehabilitation interventions in stroke with equal access for all. This requires integrating expertise in motor learning, sensors and robotics, data science and AI, and biomechanical modelling. In addition, it also requires understanding the complexity of settings and attitudes and beliefs of both clinicians and patients, including vulnerable patients and patients with a low socioeconomic status (SES).

To reach beyond the initial implementation phase we will integrate four main pillars:

1. identifying requirements of user-centred minimally supervised treatment that also target lower SES and vulnerable patients,
2. optimizing the effectiveness of (remote) motor learning programmes by exploiting the relation between motor learning, motor capabilities and daily life performance and behaviours,
3. engineering home training environments that leverage novel ultra-low-cost technological solutions that are minimalistic and easy to use to promote the transfer of trained skills to activities of daily living,
4. developing unobtrusive remote monitoring and feedback based on sensor fusion and modelling to quantify patient performance in their natural environment and to ensure their safety.

The future impact goes beyond the stroke population and will include other neurological and musculoskeletal diseases, like multiple sclerosis, spinal cord injury, Parkinson's disease and osteoarthritis, which limit mobility

1.c Public Summary

Yearly in the Netherlands between 40,000 and 45,000 patients suffer a stroke. 80% subsequently suffer from motor impairments such as paralysis or spasticity, needing help with daily activities and in participating in social activities and work. This leads to loneliness, psychological problems and loss of quality of life. Motor rehabilitation is crucial after stroke, but it is also labor-intensive and not particularly accessible. Further, the demand for care after stroke will increase by 35% by 2035 alone [1,2,3].

Now that the limits of the financial, personnel and social sustainability of healthcare are approaching, rehabilitation needs to become more effective and efficient. Technology can contribute by facilitating rehabilitation in one's own home environment. Patients will become less dependent on one-on-one guidance from a therapist, have to travel less to rehabilitation centers and will be more independent in their planning. Ease of use, accessibility and acceptance are important challenges of new technology. Importantly, vulnerable groups may not benefit from it, increasing existing health differences.

The **Convergence Human Mobility Center (CHMC)** brings together technicians, clinical experts, socio-medical researchers, patients and industrial partners. Together we will build on technology-based interventions for minimally supervised motor rehabilitation at home and under minimal supervision of a therapist in the rehabilitation center. Motion sensors inform patients and their therapists about recovery and progress of treatment. Home exercise will be supported by easy-to-use robots with challenging (virtual) exercise programmes tailored to the patient. These developments are emphatically done in collaboration with vulnerable groups so that they too benefit fully from this new form of rehabilitation. The future impact goes beyond the stroke population and will include diseases like multiple sclerosis, spinal cord injury, Parkinson's disease and osteoarthritis.

2. Relevance to Convergence Health & Technology mission

2.a Contribution to H&T mission and fit with H&T theme(s)

The care for people who have suffered a stroke constitutes the second-highest annual cost in healthcare (after dementia). Post-stroke motor rehabilitation is labor-intensive and costly as in neurological disorders that involve motor disability such as spinal cord injury (200 patients/year) and multiple sclerosis (17,000 patients/year). Currently, over 1.5 million patients suffer from (neurological) motor impairments, of which around 250,000 are wheelchair dependent and 500,000 are socially isolated due to their impairment. Patients are currently required to travel to rehabilitation centres to receive labour-intensive one-on-one therapy. Moreover, as our society ages, the number of stroke patients in the Netherlands is expected to increase by 34% between 2015 and 2035 [3], increasing the pressure on our healthcare system as available resources in rehabilitation will not match the anticipated growth in the number of people needing care.

TU Delft, Erasmus MC and Erasmus University Rotterdam are joining forces with Rijndam rehabilitation and Laurens geriatric rehabilitation in a Convergence Human Mobility Center (CHMC). By integrating expertise, knowledge and tools from (socio-) medical science, neuroscience and engineering we will improve the efficacy of **technology-based minimally supervised and tailored motor rehabilitation** paving the way to minimally supervised motor rehabilitation in which clinicians provide healthcare remotely from patients.

This will replace labor-intensive and face-to-face interventions, help patients to adhere to treatment protocols by providing more training intensity and continuity of care, and by

eliminating the need to travel. Long-term use will be enhanced by identifying influential factors in its uptake and by developing strategies to address barriers and leverage identified facilitators. We will avoid a “one design for all” approach to prevent health inequality. The outcomes will also be relevant for other diseases like multiple sclerosis, spinal cord injury and Parkinson’s disease.

As such, the CHMC will address three of the H&T missions:

- Life-long health and well-being, from pre-birth to end of life
- Socio-economic equality in health(care)
- Individually tailored precise medical treatment for all

2.b Synergy between disciplines

Technology-based minimally supervised motor rehabilitation represents a paradigm shift from therapists guiding patients in therapy towards patients managing their own minimally supervised therapy with ultra-low-cost interactive technology. This requires an interdisciplinary effort integrating expertise and tools from (socio-) medical science, neuroscience, engineering and commercial partners. This will transform the traditional rehabilitation processes into a connected system encompassing the entire health journey, integrating inpatient and outpatient rehabilitation, formal and informal care with multi-layered data and neurorobotics. Merging interrelated areas of knowledge and distilling them into minimalistic solutions will contribute to data-driven, resilient, inclusive and affordable healthcare systems. As the adaption of modern technology is challenging for vulnerable people with less resilience, equality in healthcare utilization is one of the spearheads of the CHMC flagship.

The CHMC brings together socio-medical expertise (EUR: Cramm), medical sciences (Erasmus MC: Ribbers, Frens, Bussmann, Selles) and engineering and modelling knowledge (TU Delft: Vallery, Marchal-Crespo, Seth, Kok)). We aim to create person-centered minimally supervised treatments for (lower educated) patients, exploiting the relation between motor learning, motor capabilities, motivation and boosting retention of learning. We will develop technological solutions that are minimalistic, ultra-low-cost and easy to use, and accurate remote monitoring of motor behaviour with unobtrusive measurements as predefined deliverables.

All stakeholders are involved in the development, evaluation, dissemination and utilization process. All PhD’s students will be supervised by two promoters of different institutes. Two postdocs will function as liaisons through all the work packages, one with a focus on patient participation and co-creation of care and a second one with a focus on technological development. Rijndam (medical rehabilitation) and Laurens (geriatric rehabilitation) are strong clinical partners with direct-to-patient (DTP) experience and will be involved in all stages of the project. So will Pharos, the national expert center on health disparities.

The technology must be *especially designed for* and *co-created with* patients, in such a way that its application leads to enhanced participation, better treatment and better health, especially for vulnerable patients who benefit less from current care delivery. The two postdocs will maintain mutual insights on how patient participation and co-creation of care and technological innovations interact.

2.c Long term sustainability plan

Feasibility

To address the opportunities and risks associated with technology-based minimally supervised motor rehabilitation, the values and perspectives of all potential users will be integrated upfront. Patients, clinicians and innovation managers of Rijndam and Laurens and private sector companies will be part of the research team. We will facilitate the long-term sustainability of our inventions by rigorously selecting candidate solutions based on viable and sustainable business models. Thus, private industry and the economy of scale will be involved from the beginning of the program to make the technology fit the patient, to reduce costs by mass production of sensors and robotic devices and to deliver maintenance, training and data management services.

Definition of success

With a focus on motor learning and patients' motivation, the (re)design of usable ultra-low-cost devices, creation of advanced sensor data-driven models, and technology-supported motivating (at-home) treatment will be considered successful when:

- a long-lasting lively community is created also involving students from the three institutions;
- patients feel motivated through the person-centered solutions and outcomes improve (e.g., social participation, self-management, well-being);
- the effectiveness of the person-centered solutions is more effective among lower educated patients than current solutions;
- outcomes from motor rehabilitation programmes with minimal and remote supervision are as good or preferably better than traditional one-on-one supervised therapy settings;
- at-home monitoring and data-driven models provide insights that translate to improved outcomes;
- the skills gained during robotic rehabilitation transfer to function in activities of daily living;
- co-creation workshops are structurally embedded within the four programme's research themes;
- the research approach spreads to adjacent neuromotor diagnoses and clinical fields.

Sustainability

To ensure the sustainability of the innovations, evaluation beyond the preliminary implementation stage is required. We propose strategies for both workforce and system changes for new technology-based healthcare practices to become incorporated into routine services.

Four main themes are distinguished:

- [1] perceptions of minimally supervised systems and technology;
- [2] organizational and cultural environment;
- [3] processes of clinical care, and
- [4] benefits of utilization for diverse groups of patients and clinicians compared to pre-implementation.

This approach targets long-term sustainability, covering the organizational perspective, clinician perceptions, and patient's factors. Strategies will be developed along subsequent implementation phases including ease of use of the technology (pre-implementation), system changes in practice (early implementation), data collection and individualized feedback modules (long-term implementation).

We will pay special attention to the incorporation of acquired knowledge into the curriculum of students of Erasmus MC, TU Delft and EUR (next generation of innovators).

Risks

Patient inclusion and organizational change are essential to successfully proceed in this CHMC flagship. In minimally supervised motor rehabilitation the DTP-approach is essential. Both Rijndam and Laurens had to reorganize traditional rehabilitation programmes during the COVID-19 pandemic and already have integrated decentralized clinical care in clinical routines and have made it a spearhead in the innovation portfolio. As such Rijndam and Laurens are strong clinical partners with DTP experience. This mitigates the risk of a lack of patient inclusion and failed implementation in clinical care.

Although considered as the gold standard, randomized controlled trials (RCTs) have limitations preventing their generalizability to real-world practice. Because of specific inclusion and exclusion criteria, RCTs do not account for the broader population managed in routine clinical practice and its specificities, e.g., vulnerable populations, ethnic differences, comorbid conditions and differences in lifestyles. Real-world evidence (RWE) complements traditional clinical trial data, especially in the assessment of safety and efficacy in real-world settings. Real-world data (RWD) consist of information collected during routine clinical practice. The use of digital health data (often from devices such as digital watches) on top of patient-generated data (mostly generated from electronic health records) are increasingly being used to optimize research and development programs. This is a spearhead of the department of rehabilitation medicine with extensive experience in data-driven care pathways. This mitigates the risk that RCTs are of limited value in evaluating technological innovations.

2.d Visibility, attracting talents and new partners**Collaborative mindset**

Talent acquisition is vital to realize the interdisciplinary nature of the Convergence Human Mobility Center. Diversity brings complementary insights, skills and experiences – valuable resources that can be used to rethink primary tasks and strategies. Therefore, the integration of clinical, technical and sociomedical researchers is a strategic spearhead. In the CHMC, clinicians, technicians, movement scientists and socio-medical scientists will interact in lab meetings, research projects, grant applications and education. In joint projects, we will share PhDs, postdocs, VENI and VIDI laureates, and invest in improving collaboration between researchers at different career stages from junior to senior staff.

Talent development

From the start of their appointments, PhD students and postdocs will be stimulated to develop their skills and CVs to increase their chances of acquiring personal and other grants. In this process, they will be supported by interdisciplinary writing teams from within our core staff. Junior and senior researchers, PhD students, and sometimes guest speakers will be invited to present their research in weekly interdisciplinary lab meetings. Twice a year we will organize strategy sessions to discuss team dynamics and the progress of our research program. We will support and motivate researchers in presenting their work and in representing the CHMC Flagship and its partners in conferences and other meetings. Finally, we will develop a structure around employee training and skill development aimed at the retention of the most vital and talented people who are also likely to be a good fit with our mission goals.

Involvement of early-career researchers

We will involve PhD students and postdocs in strategic and management activities of the program. We will give them responsibility in creating an internationally renowned center of expertise. This will be beneficial to all: besides strengthening a team attitude, senior staff will gain new insights while junior researchers will improve their strategic skills. Undergraduate students will be

involved through internships and the minor program Neuro Rehabilitation (joint minor program TU Delft / Erasmus MC, running since 2019; in 2022 we will add the theme of human mobility as a common thread).

Visibility

Our **clinical** ambition is to develop user-centered technology-based interventions for minimally supervised motor rehabilitation for personalized rehabilitation of neurologically impaired patients. The impact is expected to spread from stroke to other diagnoses such as spinal cord injury and multiple sclerosis. Our **scientific** ambition is to be internationally recognized as a leading institute for med-tech research in **mobility** by building a research community ranging from low to higher technology readiness levels.

In July 2022 the Rehabweek will take place in Ahoy, Rotterdam. The Rehabweek is the major world congress on rehabilitation technology, welcoming over 1,500 delegates from all around the world as well as the leading med-tech companies. We plan to officially launch the Convergence Human Mobility Center during the site visit to Erasmus MC / TU Delft / Rijndam with the Convergence of Health and Technology as the central theme.

2.e Long term societal impact

Main socio-economic challenge

Neurological motor impairments enormously impact the patients' health, independent functioning and social participation with a huge socio-economic impact. To maintain the high standards and the accessibility of health care in the future, treatment costs need to be reduced. Interventions for motor impairments are resource-intensive and provided on a one-on-one basis between clinician and patient. As intensity of practice is the major determinant of outcome, limited resources cause limited outcomes. Remote patient-tailored interventions with a clear objective and quantifiable impact are urgently needed.

Impact

This CHMC flagship aims at developing and applying innovative signal-analytical techniques, and relating original methods from the areas of user-centred design and "applicability for all" to a personalized minimally supervised rehabilitation perspective. This will facilitate the transition to using technology in at-home treatment, contributing to more cost-effective motor rehabilitation. The impact will expand from stroke to other impairments requiring motor rehabilitation such as spinal cord injury, multiple sclerosis, cerebral palsy and Parkinson's disease. Investing in effective and usable technology will lead to long-term cost savings. Furthermore, this project has the potential to reduce persistent socio-economic health inequalities by truly tailoring innovations to the needs of low SES patients and offering personalized learning strategies.

Impact pathways

The pathway to groundbreaking science and innovation is by establishing a translational research environment where clinicians, scientists, engineers and companies collaborate in achieving science-driven breakthroughs in human mobility. Rehabilitation clinics Rijndam, Laurens and various industry partners will participate and contribute from the start.

Pharos (the Dutch Centre of Expertise on Health Disparities) will help create new knowledge on how to include low-SES patients as co-producers and co-creators in the project. This knowledge will be disseminated via tools / protocols on how to include disadvantaged groups in research and practice.

Expected TRLs and necessary partners

The challenge of the CHMC is to think big (vision) but start small enough with minimal viable products and fail quickly (test and learn) to scale fast. Starting small and demonstrating progress is key to building credibility and expanding a network of interested stakeholders. Scaling

fast becomes more likely with the ability to leverage the proof point of a successful real-world experiment.

The researchers of the three faculties have varying backgrounds and so do the TRL's of the solutions and products they are working on. Dr. Marchal-Crespo and Prof. Vallery do translational research together with clinicians and companies (e.g., Motek, NL and Hocoma, CH) in co-creating robotic/VR solutions for upper and lower limbs neurorehabilitation (TRL 7-8). Musculoskeletal modeling and sensor fusion methods are embedded in open source (TRL 8) and commercially available software and sensor systems (TRL 9). Algorithms for coupling sensor data through musculoskeletal modeling are being developed (TRL 2-3) and provide an opportunity to extract patients' information that is therapeutically actionable by low-cost devices (TRL 7-8).

The CHMC flagship offers the perspective to bring the technology from TR levels 5-6 towards clinical implementation (TRL 7-9). This is warranted by clinical partners. Private industry and the economy of scale are necessary to make the technology fit the patient and to reduce costs by mass production of sensors and robotic devices and delivering maintenance, training, and data management services. Typically, the private industry has more expertise in advertising and promotions and getting innovations at TRL 9 warranting early involvement in the process.

3. Flagship proposal (25%)

3.a Ambitions

The CHMC aims to improve outcomes of motor rehabilitation through minimally supervised therapy, preferably at home, and supported with wearable sensing, ultra-low-cost technology and personalized learning strategies. In this proposal, fundamental insights in neural plasticity, neurorobotics, sensing technology and AI are integrated with strategies to support patients in self-management by an interdisciplinary experienced research team.

Our innovations include the (re)design of affordable easy-to-use (robotic) devices that provide realistic sensorimotor training that allows user-friendly therapy personalization.

We propose groundbreaking technologies to observe the internal state of the patient and estimate motor-control variables from unobtrusive external sensor measurements. We will use the latest modelling, genetic and electrophysiological insights to optimize the learning process into a personalized plan. These will be combined with motivational strategies empowering self-management, especially in people with a lower SES background who will be involved in the process of co-creation.

Our solutions will broaden the possibilities of using the latest technology at home, as they are tailored to patients with a broad range of disability levels and with the highest health disparities. The latter requires built-in flexibility in the application so that our research can transfer to other applications in movement training, e.g., other neurological and musculoskeletal disorders.

3.b Aims and feasibility

Program aim and objectives

To enhance the effectiveness and acceptability of minimally supervised rehabilitation with equal access for all we will meet four key objectives:

1. **Boost motivation and self-management.** We will boost motivation and self-management during home treatment using person-centered innovations.
2. **Boost motor training effectiveness.** We will exploit electrophysiology, genetics and motor behaviour of neurologic patients to develop optimal rehabilitation strategies for personalized at-home rehabilitation.
3. **Boost accessibility.** We will design, develop and evaluate accessible, motivating and acceptable training environments. We will boost accessibility by rigorously selecting candidate solutions based on viable and sustainable business models.
4. **Boost assessment.** We will develop monitoring and modeling solutions to ensure patient safety, extract meaningful outcome measures of motivation and movement quality from heterogeneous sensor data and provide real-time feedback in unsupervised settings.

Team and talent

We will hire 7 PhD students and 2 postdocs, co-supervised by at least two promotor from different institutions, crossing disciplinary boundaries. We have access to talented students and researchers for instance from the interdisciplinary Master's programmes, e.g., the MSc Technical Medicine, which trains students in both medicine and technology. In the BSc programme in Health Policy & Management at EUR, students acquire the ability to think in a multidisciplinary way to understand the complexity of policy and management issues in healthcare.

Expertise

We build on extensive expertise in the understanding, prevention and treatment of mobility limitations (see also section 4).

Erasmus MC, Ribbers, Bussmann, Selles and Frens' groups have extensive expertise on the upper extremity on prediction of recovery, precision diagnostics with neurorobotics, and exploiting and stimulating neuroplasticity.

TU Delft, Marchal-Crespo and Vallery co-create robotic/VR solutions for upper/lower limbs neurorehabilitation together with clinical and industry partners. Seth, Kok and Zgonnikov have deep expertise in musculoskeletal modelling, sensor fusion, and decision making modelling.

EUR, Cramm investigates innovative healthcare solutions for vulnerable patient populations. Her participation will merge the medico-technical domain of motor impairments with the social approach of inclusiveness.

Funding plan

The system development, data collection, salary of PhD students/postdocs, supervision and project management will require 2M€ funding from Convergence in addition to the in-kind contributions from the three Convergence partners. This amount will be matched with several co-financing sources. Rijndam's contribution involves clinical researchers and a liaison with its Living Labs. Envisioned co-financing from industrial collaborators includes companies such as Hocoma, SenseGlove, CleVR, 2M Engineering, Hankamp Rehab, and Lode Holding.

Our consortium will be in an excellent position to submit proposals to national (ZonMw, NWO, topsector LSH) and external (e.g., Netherlands Brain Foundation, EU ERC) grant organizations, since we share similar research agendas (see section 3.d). Our research will unlock other applications, broadening the possibilities of funding.

We will include several ongoing projects in the program a.o.:

- Armcoach4Stroke (TUD, EMC) ZonMw, Hartstichting
- Comet and FallCatcher (TUD, EMC, SRA lab), NIDILRR (USA)

- Senseglove (Rijndam, EMC, CleVR, Senseglove), IQ
- Developing innovative care pathways (EUR, EMC, Rijndam), ZonMw
- 'Cloud Walker' project. 'Open Mind Convergence' (Rijndam, EMC, TUD)

We will submit the following consortium proposals in 2022/2023:

- Access to care in the living environment (NWO, 2022).
- Towards a tailored monitoring and training tool for at-home rehabilitation (TKI 2022)
- Advancing musculoskeletal rehabilitation through biomechanics-aware robotics (NWO_OTP 2022)
- Sensor-based simulation of human movement (NWO-M2, 2023).

We also plan to guarantee the continuation and expansion of the CHMC by submitting broader consortium grants, e.g., NWO-Perspective or Gravitation. Our international expansion will be guaranteed by applying for European grants from year 2. We were already successful in applying for US grants with Shirley Ryan Ability Lab and will continue this cooperation.

Further growth and sustainability

We will create and consolidate interdisciplinary collaborations with other departments, institutions and industrial partners in the Netherlands and beyond. Our intended program coordinator dr. Sluijter has a strong track record in strengthening and expanding public private partnerships (business director Neurocontrol, director Medical Delta, responsible for preparing trade mission rehabilitation 2015 with the signature of the MOU with Shirley Ryan Ability Lab).

In 2021, interviews were held with department heads of clinical departments (EMC) who wholeheartedly welcome the start of a CHMC and are interested in joining. Once the foundation has been laid, and the time has come expand to other diagnosis groups, it will be the right moment to involve them.

Risk mitigation

- 1. Recruiting patients (low). Mitigation.** Rijndam and Laurens will be involved as clinical partners (Rijndam alone: 600–800 stroke patients/year).
- 2. Our solutions might not show enough relevance to rehabilitation (medium). Mitigation.** There is ample neurophysiological evidence supporting the flagship concept.
- 3. Lack of users' acceptance of the developed technology (medium). Mitigation.** We will create ultra-low-cost and very easy-to-use solutions. We have experience in co-creating clinically-driven designs to enhance acceptance. Co-creation will be facilitated by patient panels already formed by Rijndam.
- 4. Delayed due to METC approvals (low). Mitigation.** We have broad experience with METC approvals.

3.c Work plan



WP1: Boosting motivation and self-management**Lead: EUR-Cramm. Co-lead: EMC-Ribbers****Objective**

To explore user-centred minimally supervised treatment for lower SES and vulnerable patients.

Description

This WP aims at boosting motivation and self-management in minimally supervised training settings among patients with mobility problems. Personal interaction, encouragement and positive social pressure of the therapist are missing at home, which often makes patients slack during home treatment, especially in lower educated patients. Current treatment is scarcely adapted to the preferences and abilities of lower-educated patients and the application of technology risks increasing this gap between lower and higher educated people. We aim to personalize rehabilitation technologies for and together with lower-educated patients. We expect that they can be reached when rehabilitation technologies are tailored to their specific needs and delivered in the comfort of their own home.

Tasks

T1.1: Identify requirements of person-centered home treatment for (lower educated) patients with mobility problems.

T1.2: Predict the impact of person-centred innovations on outcomes for (lower educated) patients (e.g. motivation, self-management).

Team

PhD supervised by EUR-Cramm and EMC-Ribbers (T1.1-2).

WP2: Boosting effectiveness**Lead: EMC-Frens. Co-lead: TUD-Marchal-Crespo****Objective**

To optimize the effectiveness of (remote) motor learning programmes by exploiting the relation between motor learning, motor capabilities and daily life performance and behaviours.

Description

This WP aims at optimal and personalized motor learning exploiting electrophysiology, genetics and motor behavior based on a few parameters such as genetic polymorphisms, variations in electrophysiological parameters, and variability in baseline motor behavior. This is more efficient than a one-size-fits-all approach. We devise methods that stimulate motor learning while discouraging strategic behavior and will apply these insights to greatly boost the effectiveness of rehabilitation, especially in minimally supervised environments.

Tasks

T2.1: Identify the relation between motor learning, motor capabilities and motivation to boost therapy effectiveness.

T2.2: Develop protocols that minimize strategic behaviour to boost retention and generalization of what is learned during rehabilitation.

Team

PhD supervised by EMC-Frens and TUD-Vallery (T2.1).

PhD supervised by TUD-Marchal-Crespo and EUR-Cramm (T2.2).

WP3: Boosting accessibility**Lead: TUD-Marchal-Crespo. Co-lead: TUD-Vallery and EMC-Ribbers****Objective**

To engineer home training environments that leverage the use of novel technological solutions and are minimalistic, ultra-low-cost and easy to use.

Description

We will design, develop and evaluate accessible, motivating and acceptable training environments minimalistic in hardware, ultra-low-cost and easy to understand/use such as simple input devices, virtual (VR) or augmented (AR) reality off-the-shelf head-mounted displays (HMDs) and minimally actuated robotic devices. Rigorous selection of candidate solutions based on viable and sustainable business models will facilitate further dissemination and boost accessibility.

Tasks

T3.1: Develop minimally-actuated portable robotic haptic devices to stimulate sensorimotor training while offering an effortless setup. Combine robots with immersive VR/AR visualizations to naturalistically simulate real/imaginary activities of daily living that can be adapted to the patients' needs to enhance their motivation and reduce their cognitive load.

T3.2: Investigate how easy-to-use motivational toys (such as our robot ball "Fizzy") can be adapted for marginalized adults post-stroke.

Team

PhD supervised by TUD-Marchal-Crespo and EMC-Ribbers/Bussmann (T3.1).

PhD supervised by TUD-Vallery and EUR-Cramm (T3.2).

WP4: Boosting assessment**Lead: TUD-Seth. Co-leads: EMC-Bussmann, TUD-Kok, EMC-Selles, TUD-Zgonnikov****Objective**

To develop unobtrusive remote monitoring based on sensor fusion and modelling to extract meaningful measures in the patients' natural environment.

Description

We will derive measures of motor capacity and performance from wearable sensing technology in the natural (home) environment: at-home movement assessments during standardized motor tests, during ADLs, and during at-home therapy sessions. We will build neuromechanical models to transform sensed movements and robot measured forces into metrics of central and peripheral activation, motor noise, and motor learning and use musculoskeletal models to estimate the loads in the musculoskeletal system so that risks of injury are mitigated while performance is improved.

Tasks

T4.1: Develop a data-driven decision model identifying patients' capacities, behaviours and motivation from unobtrusive measurements of patients in their natural environment.

T4.2: Develop a real-time framework for neuromusculoskeletal modeling-based sensor fusion that combines heterogeneous sensors readings to estimate the patient's state.

Team

PhD supervised by EMC-Bussmann, EUR-Cramm, and TUD-Zgonnikov (T4.1).

PhD supervised by TUD-Seth/Kok and EMC-Selles (T4.2).

Program liaisons

Two postdocs will be employed as program liaisons. One postdoc (EUR) will build and maintain insights concerning patient participation and co-creation of care, facilitate synergetic collaboration, communications and coordinate activities to make the most benefit out of the flagship. A second postdoc (TUD) will in a similar way be responsible for spreading knowledge on technical issues related to at-home technological innovations suited for vulnerable patients.

3.d Alignment of the program with relevant national and international research agendas

The CHMC targets the effective use of knowledge through co-creation, one of the strategic spearheads of **NWO**. In addition, it aims to valorize knowledge in everyday practice, aligning with the focus area identified by the **Health Council of the Netherlands** on modern technologies that help promote health and stimulate equality in healthcare. The CHMC will concentrate on signaling bottlenecks in knowledge valorization and boost efforts to accelerate it. Vulnerable people with less resilience are a particular focus area in the CHMC flagship.

Top Sector Life Sciences & Health supports innovative research and development realized by public-private partnerships, helping consortia consisting of research organizations, knowledge institutes, companies and health foundations realize their innovative ideas. The partners involved in the CHMC already have extensive experience with PPPs in a broad network of med-tech companies. Decreasing health inequalities between socioeconomic groups is in the central mission of LSH, which is also addressed in the CHMC. LSH has defined four underlying aims: (1) changes to lifestyle, (2) changing the living environment, (3) offering care in the right place, and (4) better perspectives for people with lifelong diseases/disabilities. The CHMC addresses all these aims.

European research strategies: open call Cluster 1 Health work programme is planned for 2023-2024 on augmented medicine, including medical imaging, neuromechanical modeling and robotics. This will support the development and integration of technologies, i.e., neurorobotics, imaging data visualization tools and VR/AR, to advance safety and efficiency in diagnostics and medical interventions.

4. Consortium (25%) max 2000 words for 4.a to 4.d**4.a Track record of the consortium members**

<i>Name</i>	Gerard Ribbers (main Lead)
<i>Track record research</i>	Full professor Neurorehabilitation and Head of the Department of Rehabilitation Medicine at Erasmus MC: over 160 peer-reviewed papers, H-index 42, i10-index 92, 5513 citations (Google Scholar), 9 finished PhD thesis and 7 ongoing, > 6m€ funding.
<i>Fit expertise-program</i>	During 25 years Ribbers' clinical tasks have focused on patients with severe cognitive, behavioural and linguistic disorders due to acquired brain injury. He was appointed Professor of Neurorehabilitation in 2012 and Chair of the Department Rehabilitation Medicine in 2018.
<i>Connection to external partners</i>	Member steering committee Innovative Medical Devices (IMDI) - <i>Neurocontrol</i> with clinical leadership on Neurostimulation. Leader of Medical Delta theme "Innovative Rehabilitation Technology. President Executive Committee of the International Brain Injury Association (2015-2016). Board member Rotterdam Stroke Service 1999-2018. Member International Consortium of Health Outcomes Measurement. Steering Committee of the Stroke Standard set (since 2014). Host RehabWeek 2022 in Rotterdam, the Netherlands.

<i>Name</i>	Jane Murray Cramm (PI and WP lead)
<i>Track record research</i>	Full Professor at the Erasmus School of Health Policy & Management, EUR. Published over 130 peer-reviewed articles, H-index 38, i10 index 102, 5454 citations (Google Scholar), co-supervisor 200+ Master's theses, supervised 6 PhD students, 6 ongoing, 3.500+ k€ funding.
<i>Fit expertise-program</i>	Research on person-centred care and diversity is a core component of Cramm's work. This project allows the investigation of personalized innovations in the rehabilitation setting and how this may help to reduce socio-economic health inequalities using a multidisciplinary and transdisciplinary perspective.
<i>Connection to external partners</i>	Member of the central mission team Topsector theme Health (Kennis- en innovatieagenda 2020-2023 Gezondheid & Zorg together with Pharos).

<i>Name</i>	Laura Marchal-Crespo (PI and WP lead)
<i>Track record research</i>	VIDI laureate and associate Professor at TU Delft (0.8 FTE), associated with the ARTORG Center for Biomedical Engineering Research, University of Bern (0,2 FTE). H-index 20, i10-index 33, 2904 citations (Google Scholar), > 65 peer-reviewed papers, co-supervised 39 Master's theses and supervised 10 PhD students (5 ongoing), 5.000+ k€ funding including Swiss national foundation professorship and NWO-VIDI.
<i>Fit expertise-program</i>	Research focus on human-machine interfaces and biological motor learning, in particular the use of robotic devices and VR for the diagnosis, treatment and rehabilitation of neurological patients. She develops technology to record, analyze, adapt and respond to patients' motor, sensory and cognitive needs to maximize recovery.
<i>Connection to external partners</i>	Strong industrial (e.g., Hocoma, Force Dimension, Switzerland and SenseGlove, NL), clinical (e.g., University Hospital Bern Inselspital, Switzerland) and academic partners in the rehabilitation and human-robot interaction fields.

<i>Name</i>	Maarten Frens (WP lead)
<i>Track record research</i>	Full professor in Systems Physiology at the Neuroscience department, and Vice-Dean of Education at Erasmus MC. Published 100 peer-reviewed papers. H-index 35, i10-index 179 (Google Scholar), supervised 25 PhD students (2 cum laude, 5 ongoing), 8.000+ k€, including NWO-VIDI, HFSP, Marie Curie FP7 ITN.
<i>Fit expertise-program</i>	Frens' research is on the neural mechanisms involved in motor learning involving behavioural studies in humans and animals. In collaboration with the Department of Rehabilitation this expertise has been translated to bedside application.
<i>Connection to external partners</i>	Involved in several initiatives to connect academia and the city of Rotterdam, including programmes for first-generation students and the Cultuurencampus in Rotterdam Zuid.

<i>Name</i>	Ajay Seth (WP Lead)
<i>Track record research</i>	Assistant professor biomedical engineering, H-index 19, i10 index 21 with 28 peer-reviewed journal publications. Total citations: 4065 for an average of 145 cites/publication, PhD supervisor of 3 PhD students at TUD, exam committee of 2 PhDs (Twente and TUD), and successfully mentored 6 PhDs at Stanford University. Supervisor of over 25 MSc-thesis and BSc-honors projects, >220K EUR (2020-21) funding.
<i>Fit expertise-program</i>	Expert in neuromuscular and musculoskeletal modeling and computer simulation and was chief architect of OpenSim—the most widely used biomechanical simulator in the world for human (and animal) movement research.
<i>Connection to external partners</i>	Dr. Seth has an international network of collaborators and OpenSim users ranging from Stanford University to the University of Twente. Industrial users range from Silicon Valley tech startups to multinational orthopedic implant companies.

4.b Expertise of the consortium members in relation to the aims and objectives of the program

The aims of the programme require not only technological and clinical expertise but also a deep understanding of the relationships between person-centred technology-based transitions, motivation and self-management abilities of patients to maintain overall well-being and promote social participation. For a self-management programme to work well, it must include an educational/behaviour change dimension and functional links with the healthcare system.

For this reason, the expertise present at TUD, EMC and EUR in understanding, preventing and treating mobility limitations will be converged, specifically the social and medical perspectives on disability. To enable co-creation with patients, Rijndam and Laurens are on board as partners, and as clinical expertise and test centres. The multidisciplinary of the programme is reflected by the main applicants. VIDI laureate Heike Vallery, full professor TUD and honorary professor EMC, is a mechanical engineer, specialized in bipedal locomotion and human-robot haptic interaction. Manon Kok, assistant professor at the TUD Center for Systems and Control, is specifically interested in sensor fusion using inertial sensors and magnetometers for (human) motion estimation. Her research focuses on developing new algorithms to extract more information from the available sensor data. Arkady Zgonnikov, assistant professor TUD, bridges the fields of human motor control, decision making, and artificial intelligence. His current research focuses on computational and data-driven modeling of human behavior in human-AI and human-robot interactions. Hans Bussmann, associate Professor, Rehabilitation Medicine EMC, aims at understanding and improving physical behaviour of people with chronic conditions, and to apply this in rehabilitation care. His focus is on the use of wearable technology in the "at home" situation. Ruud Selles, professor EMC, Rehabilitation Medicine and Plastic Surgery, has a long track record in upper extremity research and treatment in neurorehabilitation. He holds a chair on individualized treatment of the upper extremity with the ambition to use innovative research approaches to find better ways to treat disorders of the upper extremities and extends these innovative approaches to other research areas in rehabilitation medicine. Erik Grauwmeijer is MD and clinical researcher. His expertise both in clinical care and research is acquired brain injury. Ruben Regtershot, postdoc rehabilitation medicine, his research is on wearable movement sensors: algorithm development, data analysis and machine learning. All researchers in this flagship are highly knowledgeable in their fields of expertise and experienced in cross-disciplinary collaboration. The Convergence H&T grant will allow cross-fertilization and an interdisciplinary approach to achieve impact and lay the foundations for long-term collaboration.

For description of leads Ribbers, Cramm, Marchal-Crespo, Frens and Seth: see section 4.a.

4.c Involvement of partners

Rijndam rehabilitation: one of the largest rehabilitation centres in the Netherlands (>800 employees, >16,000 patients, 16 locations). Motor rehabilitation after neurotrauma (stroke, spinal cord injury) is a clinical spearhead. Their ambition is to integrate innovative rehabilitation technology in clinical practice and to serve as a clinical expert and living lab. Clinical researchers participate in research of the Department of Rehabilitation Medicine. Rijndam adds physiatrists, a physical therapists, a Living lab coordinator and innovation manager to the team.

Laurens geriatric rehabilitation: one of the largest skilled nursing facilities in the Netherlands (3,400 FTE, >3,000 inpatient beds, 2,400 clients in outpatient treatment). Specialist in geriatric rehabilitation after surgery, accident or illness such as after stroke orthopedic surgery, lung disease oncology and heart failure with three complete care locations. The quality management system of Laurens is built upon routine outcome and experience measures.

Rijndam and Laurens are planning to move to the EMC campus to participate in the Convergence of H&T initiative and join forces in research.

Hocoma: global market leader for the development, manufacturing, and marketing of robotic and sensor-based devices for movement therapy. Hocoma develops innovative therapy solutions working closely with leading clinics and research centers. Prof. Vallery and Dr. Marchal Crespo have been collaborating with Hocoma since 2010 in several granted projects for upper and lower limbs robotic/VR-based rehabilitation. They will support by providing advice -e.g., in business opportunities and technology transfer, and medical device development practice such as risk management and product certification- and research collaborations -e.g., promoting exchange of our researchers/students with Hocoma's Research and Development teams.

Hankamp Rehab: distributor of an extensive range of rehabilitation aids for patients with acquired brain injuries (CVA, traumatic brain injury, MS, etc.). In addition, Hankamp is developing new innovative rehabilitation products and assists partners and patients with advice and assistance. Hankamp works with EMC, TUD and Rijndam since 2013.

2M Engineering helps clients develop their innovative sensor (hardware) solutions. They analyze requirements, find the most suitable sensors, define and prove concepts, and can prepare for industrialization and prototyping. Their integrated medical certification capabilities make them the ideal partner for med-tech product developers and scaleups. They already participate in several projects such as Attractif, ArmCoach4Stroke and work with Rijndam, TU Delft and Erasmus MC (Bussmann) for over 10 years.

Lode Holding: parent company of subsidiaries Lode, ProCare, aXtion, Umaco and Omnium. These companies deliver a wide range of high-quality products and services in the field of lifestyle, sport, prevention and care. Lode Holding has been working with Bussmann (EMC) and TUD in several projects for more than 10 years.

SenseGlove (spinoff TU Delft): manufacturer of high-end haptic feedback gloves that can "feel" and interact with objects in virtual reality, making it a useful tool to measure hand movements during rehabilitation. Senseglove, Rijndam, EMC and TUD successfully implemented the first prototype of the Senseglove in Rijndam and will build further on this to make at-home use possible in the future. Bram Onneweer, coordinator of the Living Lab, leads this project with SenseGlove and CleVR since 2018.

CleVR (spinoff TU Delft): creates virtual exposure therapy in VR, with a track record in GGZ, now expanding into rehabilitation. CleVR together with Rijndam, EMC and SenseGlove, creates a virtual treatment program for fine motor control of the hand, where the treatment is personalized to individual patients.

Shirley Ryan Ability Lab (SRALab) Chicago, USA: world's leading institution in serving individuals with disabilities, combines the fields of rehabilitation medicine, biomedical engineering, neuroscience, and outcomes research. The SRALab has already very successfully collaborated with Prof. Vallery and Prof. Ribbers in multiple past and present projects. Currently, in the COMET project, on wearable balance-assisting actuators, and in the FallCATCHER project on fall injury prevention.

Pharos: Dutch national centre of expertise on health disparities. Focus on:

- Sustainable improvement of the quality, effectiveness and accessibility of care and prevention for people with limited health literacy, non-western migrants and refugees.
- Reduce existing health disparities between different groups of people.

Prof. Cramm regularly works together with people from Pharos.

4.d Governance, organization and collaboration strategy of the consortium / 221 words

The core team, with Dr. Laura Marchal-Crespo, Prof. Jane Cramm and Prof. Gerard Ribbers will be supported by a strategic advisor (Agaath Sluijter). In quarterly meetings, the core team will monitor progress in the programme and discuss new funding opportunities. One of the postdocs and one of the PhDs will join the core team and serve as linking pins to the early-career community. A user committee consisting of patients, caregivers and companies will meet with the core team and work package's leads twice a year.

Postdocs and PhDs will have monthly meetings with the core team to discuss which conferences to attend, which journals to publish in, and to evaluate team dynamics and funding opportunities. They will also invite external presenters. PhDs, postdocs, and supervisors will visit the premises of the private partners regularly, especially during the design and prototyping phases. Getting to know their development processes for commercial products will facilitate technology transfer.

We will use an Agile design & development process in all work packages. We will form a team with all partners per WP, meeting biweekly for 1-2 hours. Each meeting will identify immediate goals and specific tasks (called a Sprint) to be worked upon as a team. During the team meetings, we will discuss the status and progress of our activities in relation to the Sprint goals.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table) WP leaders

A pluriannual plan will be required if your proposal is accepted

See enclosed budget template.

In this budget table the cofinancing budgets of running projects are not yet processed. We intend to add the projects mentioned in chapter 3b. to the program. In cash contributions from these projects made by private companies as well as by funding agencies are not included as such in the budget yet. This will be done in the final budget proposal as soon as the project is granted.

See attached budget proposal in Excel

II Letters of intent and letters of commitment WP leaders

On headed paper of the entity. Letters of commitment are preferable to letters of intent and letters of intent are preferable to the absence of these letters.

See attached PDF Document

Letters of commitment and / or intent of:

WP leaders

1. Prof. dr. Maarten Frens, Erasmus MC
2. Dr. Laura Marchal-Crespo, TU Delft
3. Dr. Ajay Seth, TU Delft
4. Prof. dr. Jane Cramm, EUR

Clinical partners

5. Rijndam rehabilitation
6. Laurens geriatric rehabilitation
7. Shirley Ryan Ability Lab (SRAlab)

Companies

8. Hankamp Rehab
9. SenseGlove
10. Hocoma
11. 2M Engineering
12. Lode Holding
13. CleVR
14. Pharos (will follow in next phase)

III Letter of support from department head of the main lead GR

On headed paper of the department or institution.



BY EMAIL: s.sleijfer@erasmusmc.nl

Erasmus MC

Attention to Dean, Prof. Dr. Stefan Sleijfer

Dr Molewaterplein 40

3015 GD ROTTERDAM

Extension number +31 10 7033190

E-mail g.ribbers@erasmusmc.nl

Postal Address Nc 330a

Our reference : GR/20220307

Your reference:

Date: 7 March 2022

LETTER OF COMMITMENT

for the Convergence Human Mobility Center Flagship

Dear Dean Prof. Dr. Stefan Sleijfer,

I, Gerard Ribbers, acting in my capacity of Professor and chair of the department Rehabilitation Medicine at Erasmus MC, hereby confirm that my department intends to commit to contribute in kind to the 'Convergence Human Mobility Center Flagship' project, upon confirmation of the grant award.

The department of Rehabilitation Medicine will provide in-kind contribution consisting of hours from myself as main lead, Agaath Sluiter as project coordinator, Ruud Selles as workpackage lead and Hans Bussmann as co-workpackage lead. These hours will be invested as in managing and developing the project, time spend on guiding PhD students and time spend on working on new applications (e.g. consortium grants together with this flagship team for instance NWO, ZonMW, TKI, European programs).

Sincerely,

Prof. G.M. Ribbers, MD PhD

Head of the Department of Rehabilitation Medicine Erasmus MC

Postal Address

P.O. box 2040
3000 CA Rotterdam

Visitors Address

Dr. Molewaterplein 40
3015 GD Rotterdam

Garage

Westzeedijk 361
3015 AA Rotterdam

Head of department

Prof. Dr. G.M. Ribbers

www.erasmusmc.nl

IV References

Al Borno M, O'Day J, Ibarra V, Dunne J, Seth A, Habib A, ... Delp, SL. OpenSense: An open-source toolbox for inertial-measurement-unit-based measurement of lower extremity kinematics over long durations. *Journal of NeuroEngineering and Rehabilitation*. 2022, 19(1), 1–11. <https://doi.org/10.1186/s12984-022-01001-x>

Ballester BR, Ward NS, Brander F, Maier M, Kelly K, Verschure P. Relationship between intensity and recovery in post-stroke rehabilitation: a retrospective analysis. *J Neurol Neurosurg Psychiatry*. 2021.

Basalp E, Wolf P, Marchal-Crespo L. Haptic training: Which types facilitate (re)learning of which motor task and for whom Answers by a review. *IEEE Transactions on Haptics*, 2021.

van den Bos GA, Smits JP, Westert GP, van Straten A. Socioeconomic variations in the course of stroke: unequal health outcomes, equal care? *J Epidemiol Community Health*. 2002;56:943–948. doi: 10.1136/jech.56.12.943.

Braakhuis HEM, Berger MAM, Bussmann JBJ. Effectiveness of healthcare interventions using objective feedback on physical activity: A systematic review and meta-analysis. *J Rehabil Med*. 2019;51(3):151-9.

Braakhuis HEM, Berger MAM, Regterschot RGRH, van Wegen EEH, Selles RW, Ribbers GM, Bussmann JBJ; PROFITS research group. Physical activity dimensions after stroke: patterns and relation with lower limb motor function. *J Neuroeng Rehabil*. 2021 Dec 11;18(1):171. doi: 10.1186/s12984-021-00960-x. PMID: 34895265; PMCID: PMC8666008.

Braakhuis HEM, Berger MAM, Bussmann JBJ. Effectiveness of healthcare interventions using objective feedback on physical activity: A systematic review and meta-analysis. *J Rehabil Med*. 2019;51(3):151-9.

van der Cruisen J, Manoochehri M, Jonker ZD, Andrinopoulou ER, Frens MA, Ribbers GM, Schouten AC, Selles RW. Theta but not beta power is positively associated with better explicit motor task learning. *Neuroimage*. 2021 Oct 15;240:118373.

van der Cruisen J, Manoochehri M, Jonker ZD, Andrinopoulou ER, Frens MA, Ribbers GM, Schouten AC, Selles RW. Theta but not beta power is positively associated with better explicit motor task learning. *Neuroimage*. 2021 Oct 15;240:118373. doi: 10.1016/j.neuroimage.2021.118373. Epub 2021 Jul 8. PMID: 34246767.

Daly JJ, McCabe JP, Holcomb J, Monkiewicz M, Gansen J, Pundik S. Long-Dose Intensive Therapy Is Necessary for Strong, Clinically Significant, Upper Limb Functional Gains and Retained Gains in Severe/Moderate Chronic Stroke. *Neurorehabil Neural Repair*. 2019;33(7):523-37.

van Dijk MP, Kok M, Berger MAM, Hoozemans MJM, Veeger DHEJ. Machine Learning to Improve Orientation Estimation in Sports Situations Challenging for Inertial Sensor Use. *Front Sports Act Living*. 2021 Aug 3;3:670263. doi: 10.3389/fspor.2021.670263. PMID: 34414370; PMCID: PMC8369156.

Faber JS, Al-Dhahir I, Reijnders T, Chavannes NH, Evers AWM, Kraal JJ et al. Attitudes Toward Health, Healthcare, and eHealth of People With a Low Socioeconomic Status: A Community-Based Participatory Approach. *Front Digit Health* 2021 Jul 8;3:690182. doi: 10.3389

Hartstichting [Available from: <https://www.hartstichting.nl/hart-en-vaatziekten/feiten-en-cijfers-hart-en-vaatziekten>]

Held JPO, Luft AR, Veerbeek JM. Encouragement-Induced Real-World Upper Limb Use after Stroke by a Tracking and Feedback Device: A Study Protocol for a Multi-Center, Assessor-Blinded, Randomized Controlled Trial. *Front Neurol*. 2018 Jan 25;9:13. doi: 10.3389/

Hubbard IJ, Parsons MW, Neilson C, Carey LM. Task-specific training: evidence for and translation to clinical practice. *Occup Ther Int*. 2009;16(3-4):175-89.

Jonker ZD, van der Vliet R, Maquelin G, van der Cruisen J, Ribbers GM, Selles RW, Donchin O, Frens MA. Individual differences in error-related frontal midline theta activity during visuomotor adaptation. *Neuroimage*. 2021 Dec 15;245:118699.

Krops LA, Geertzen JHB, Horemans HLD, Bussmann JBJ, Dijkstra PU, Dekker R. Feasibility and short-term effects of Activity Coach+: a physical activity intervention in hard-to-reach people with a physical disability. *Disabil Rehabil*. 2021 Sep;43(19):2769-2778. doi: 10.1080/09638288.2020.1717650. Epub 2020 Jan 30. PMID: 31999496.

Kuipers SJ, Nieboer AP, Cramm JM. Making care more patient centered; experiences of healthcare professionals and patients with multimorbidity in the primary care setting. *BMC Fam Pract*. 2021 Apr 9;22(1):70. doi: 10.1186/s12875-021-01420-0. PMID: 33836652; PMCID: PMC8035730.

Kwakkel G, Kollen BJ. Predicting activities after stroke: what is clinically relevant? *Int J Stroke*. 2013;8(1):25-32.

Langhorne P, Coupar F, Pollock A. Motor recovery after stroke: a systematic review. *Lancet Neurol*. 2009;8(8):741-54.

Laver KE, Adey-Wakeling Z, Crotty M, Lannin NA, George S, Sherrington C. Telerehabilitation services for stroke. *Cochrane Database Syst Rev*. 2020;1(1):Cd010255.

Magwood GS, Ellis C, Nichols M, Burns SP, Jenkins C, Woodbury M, et al. Barriers and Facilitators of Stroke Recovery: Perspectives From African Americans With Stroke, Caregivers and Healthcare Professionals. *J Stroke Cerebrovasc Dis*. 2019;28(9):2506-16.

Marchal-Crespo L, Tsangaridis P, Obwegeser D, Maggioni S, Riener R. Haptic error modulation outperforms visual error amplification when learning a modified gait pattern. *Frontiers in Neuroscience*, 13:61, 2019.

Marchal-Crespo L, Rappo N, Riener R. The effectiveness of robotic training depends on motor tasks characteristics. *Experimental Brain Research*, 235(12): 3799–3816, 2017.

Marshall IJ, Wang Y, Crichton S, McKevitt C, Rudd AG, Wolfe CD. The effects of socioeconomic status on stroke risk and outcomes. *Lancet Neurol*. 2015;14:1206–1218. doi: 10.1016/S1474-4422(15)00200-8.

Naidoo P, Bouharati C, Rambiritch V et al. Real world evidence and product development: opportunities, challenges and risk mitigation. *Wien Klin Wochenschr* 2022;133 (15-16): 840-846. doi: [10.1007/s00508-021-01851-w](https://doi.org/10.1007/s00508-021-01851-w)

Nieboer AP, Hajema K, Cramm JM. Relationships of self-management abilities to loneliness among older people: a cross-sectional study. *BMC Geriatr.* 2020 May 27;20(1):184. doi: 10.1186/s12877-020-01584-x. PMID: 32460707; PMCID: PMC7254755.

Odermatt IA, Buetler KA, Wenk N, Özen Ö, Penalver-Andres J, Nef T, Mast FW, Marchal-Crespo L. Congruency of Information Rather Than Body Ownership Enhances Motor Performance in Highly Embodied Virtual Reality. *Frontiers in Neuroscience* 15, 2021.

Olivier-Pijpers VC, Cramm JM, Landman W, Nieboer AP. A multiple case study investigating changes in organizations serving residents with intellectual disabilities and challenging behaviours. *J Appl Res Intellect Disabil.* 2021 Jan;34(1):190-199. doi: 10.1111/jar.12797. Epub 2020 Sep 16. PMID: 32939925; PMCID: PMC7821192.

Otterman NM, van der Wees PJ, Bernhardt J, Kwakkel G. Physical therapists' guideline adherence on early mobilization and intensity of practice at dutch acute stroke units: a country-wide survey. *Stroke.* 2012;43(9):2395-401.

Özen Ö, Buetler KA, Marchal-Crespo L. Promoting motor variability during robotic assistance enhances motor learning of dynamic tasks. *Frontiers in Neuroscience* 14, 2020.

Prasanth H, Caban M, Keller U, Courtine G, Ijspeert A, Vallery H, von Zitzewitz J. Wearable Sensor-Based Real-Time Gait Detection: A Systematic Review. *Sensors (Basel).* 2021 Apr 13;21(8):2727.

Prendergast JM, Balvert S, Driessen T, Seth A & Peternel L. Biomechanics Aware Collaborative Robot System for Delivery of Safe Physical Therapy in Shoulder Rehabilitation. *IEEE Robotics and Automation Letters.* 2021; 6(4), 7177–7184. <https://doi.org/10.1109/LRA.2021.3097375>

The Public Health Foresight Study 2018 [Available from: <https://www.vtv2018.nl/en>] <https://www.gezondlevencheck.nl/hoe-rolstoelvriendelijk-is-nederland/>

Regterschot GRH, Bussmann JBJ, Fanchamps MHJ, Meskers CGM, Ribbers GM, Selles RW. Objectively measured arm use in daily life improves during the first 6 months poststroke: a longitudinal observational cohort study. *Journal of NeuroEngineering and Rehabilitation* volume 18, Article number: 51 (2021)

Regterschot GRH, Ribbers GM, Bussmann JBJ. Wearable Movement Sensors for Rehabilitation: From Technology to Clinical Practice. *Sensors (Basel).* 2021 Jul 12;21(14):4744. doi: 10.3390/s21144744. PMID: 34300484; PMCID: PMC8309586.

Rijksinstituut voor Volksgezondheid en Milieu [Available from: <https://www.volksgezondheidenzorg.info/onderwerp/beroerte/kosten/zorguitgaven>

Saes M, Mohamed Refai MI, van Beijnum BJF, Bussmann JBJ, Jansma EP, Veltink PH, Buurke JH, van Wegen EEH, Meskers CGM, Krakauer JW, Kwakkel G. Quantifying Quality of Reaching Movements Longitudinally Post-Stroke: A Systematic Review. *Neurorehabil Neural Repair.* 2022 Mar;36(3):183-207.

Seel T, Kok M, McGinnis RS. Inertial Sensors-Applications and Challenges in a Nutshell. *Sensors (Basel).* 2020 Oct 31;20(21):6221. doi: 10.3390/s20216221. PMID: 33142738; PMCID: PMC7662337.

Seth A, Dong M, Matias R, Delp SL. Muscle contributions to upper-extremity movement and work from a musculoskeletal model of the human shoulder. *Frontiers in Neurorobotics*. 2019 Nov, 13:1-9. <https://doi.org/10.3389/fnbot.2019.00090>

Seth A, Hicks JL, Uchida, TK, Habib A, Dembia CL, Dunne JJ, ... Delp SL. OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement. *PLoS Computational Biology*. 2018 14(7), 1–20. <https://doi.org/10.1371/journal.pcbi.1006223>

Veerbeek JM, van Wegen E, van Peppen R, van der Wees PJ, Hendriks E, Rietberg M, et al. What is the evidence for physical therapy poststroke? A systematic review and meta-analysis. *PLoS One*. 2014;9(2):e87987.

van der Velden LL, de Koff MAC, Ribbers GM, Selles RW. The diagnostic levels of evidence of instrumented devices for measuring viscoelastic joint properties and spasticity; a systematic review. *J Neuroeng Rehabil*. 2022 Feb 11;19(1):16. doi: 10.1186/s12984-022-00996-7. PMID: 35148805; PMCID: PMC8832664.

Vestjens L, Cramm JM, Nieboer AP. A cross-sectional study investigating the relationships between self-management abilities, productive patient-professional interactions, and well-being of community-dwelling frail older people. *Eur J Ageing*. 2020 Oct 23;18(3):427-437. doi: 10.1007/s10433-020-00586-3. PMID: 34483806; PMCID: PMC8377131.

van der Vliet R, Selles RW, Andrinopoulou ER, Nijland R, Ribbers GM, Frens MA, Meskers C, Kwakkel G. Predicting Upper Limb Motor Impairment Recovery after Stroke: A Mixture Model. *Ann Neurol*. 2020 Mar;87(3):383-393.

De Vries EA, Boerboom W, Van den Berg-Emons RJG, Van Kooten F, Visser-Meily JMA, Ribbers GM, Heijenbroek-Kal MH. Fatigue is Associated with Reduced Participation and Health-Related Quality of Life Five Years After Perimesencephalic Subarachnoid Haemorrhage: A Multicentre Cross-Sectional Study. *J Rehabil Med*. 2022 Feb 22. doi: 10.2340/jrm.v54.212. Epub ahead of print. PMID: 35191989.

Wang J, Geng L. Effects of Socioeconomic Status on Physical and Psychological Health: Lifestyle as a Mediator. *Int J Environ Res Public Health*. 2019;16(2).

Wenk N, Penalver-Andres J, Buetler KA, Müri RM, Marchal-Crespo L. Effect of Immersive Visualization Technologies on Cognitive Load, Motivation, Usability, and Embodiment. *Virtual Reality*, 2021.

Weygers I, Kok M, Konings M, Hallez H, De Vroey H, Claeys K. Inertial Sensor- Based Lower Limb Joint Kinematics: A Methodological Systematic Review. *Sensors (Basel)*. 2020 Jan 26;20(3):673. doi: 10.3390/s20030673. PMID: 31991862; PMCID: PMC7038336